Relations and Orderings

Math Foundations — Concept 03

1 Relations on a Set

A *binary relation* on a set A is a subset $R \subseteq A \times A$. We write $a\,R\,b$ as shorthand for $(a, b) \in R$.

Definition 1 (Reflexivity)**.** A relation R on A is *reflexive* if

$$\forall a \in A, \quad a\,R\,a.$$

Definition 2 (Symmetry)**.** A relation R on A is *symmetric* if

$$\forall a, b \in A, \quad a\,R\,b \implies b\,R\,a.$$

Definition 3 (Transitivity)**.** A relation R on A is *transitive* if

$$\forall a, b, c \in A, \quad (a\,R\,b \wedge b\,R\,c) \implies a\,R\,c.$$

Definition 4 (Antisymmetry)**.** A relation R on A is *antisymmetric* if

$$\forall a, b \in A, \quad (a\,R\,b \wedge b\,R\,a) \implies a = b.$$

Definition 5 (Equivalence relation)**.** A relation $\sim$ on A is an *equivalence relation* if it is reflexive, symmetric, and transitive. For each $a \in A$ the *equivalence class* of a is

$$[a] \;=\; \{x \in A : a \sim x\},$$

and the *quotient set* is $A/{\sim} = \{[a] : a \in A\}$.

Definition 6 (Partition)**.** A *partition* of A is a family $\mathcal{P} \subseteq \mathcal{P}(A)$ of nonempty subsets of A such that

(P1) $\bigcup_{B \in \mathcal{P}} B = A$ (covering);

(P2) $B, B' \in \mathcal{P}$ with $B \neq B' \implies B \cap B' = \emptyset$ (pairwise disjoint).

Definition 7 (Partial order)**.** A relation $\leq$ on A is a *partial order* if it is reflexive, antisymmetric, and transitive. It is a *total order* if additionally $\forall a, b \in A,\ a \leq b \vee b \leq a$.

2 Equivalence Relations Correspond to Partitions

We now prove the central structural theorem of this concept.

Lemma 1. *Let $\sim$ be an equivalence relation on A, and let $a, b \in A$. Then $[a] = [b]$ if $a \sim b$, and $[a] \cap [b] = \emptyset$ otherwise.*

Proof. Suppose $a \sim b$. If $x \in [a]$, then $a \sim x$, so by symmetry $x \sim a$, and combined with $a \sim b$ transitivity gives $x \sim b$, hence $b \sim x$ by symmetry, i.e. $x \in [b]$. The reverse inclusion is symmetric, so $[a] = [b]$.

Suppose instead $a \not\sim b$, but for contradiction take $x \in [a] \cap [b]$. Then $a \sim x$ and $b \sim x$. By symmetry $x \sim b$, and transitivity yields $a \sim b$, contradicting the hypothesis. Hence $[a] \cap [b] = \emptyset$. $\square$

Theorem 2 (Equivalence relations $\leftrightarrow$ partitions)**.** *Let A be a set. There is a bijection between the set of equivalence relations on A and the set of partitions of A, given by*

$$\Phi(\sim) \;=\; \{[a]_\sim : a \in A\}, \qquad \Psi(\mathcal{P}) \;=\; \{(a,b) \in A \times A : \exists B \in \mathcal{P},\; a,b \in B\}.$$

Proof. Step 1: $\Phi(\sim)$ is a partition. For every $a \in A$, reflexivity gives $a \in [a]$, so each class is nonempty and $\bigcup_a [a] = A$. Pairwise disjointness of distinct classes is Lemma **??**.

Step 2: $\Psi(\mathcal{P})$ is an equivalence relation. Let $\approx := \Psi(\mathcal{P})$.

- Reflexive: each $a \in A$ lies in some $B \in \mathcal{P}$ by (P1), so $a \approx a$.
- Symmetric: $a \approx b$ means $a, b \in B$ for some B; the same B witnesses $b \approx a$.
- Transitive: if $a, b \in B$ and $b, c \in B'$, then $b \in B \cap B'$, forcing $B = B'$ by (P2); hence $a, c \in B$ and $a \approx c$.

Step 3: $\Phi \circ \Psi = \mathrm{id}$. Let $\mathcal{P}$ be a partition and set $\approx = \Psi(\mathcal{P})$. Pick any $a \in A$ and let $B_a \in \mathcal{P}$ be the (unique) block containing a. The class $[a]_\approx$ consists of all x sharing some block with a, but uniqueness of the block makes this exactly B_a. Thus $\Phi(\Psi(\mathcal{P})) = \{B_a : a \in A\} = \mathcal{P}$, since every block of $\mathcal{P}$ contains at least one element.

Step 4: $\Psi \circ \Phi = \mathrm{id}$. Let $\sim$ be an equivalence relation and set $\mathcal{P} = \Phi(\sim)$. For $a, b \in A$,

$$a \,\Psi(\mathcal{P})\, b \iff \exists [c] \in \mathcal{P},\; a, b \in [c] \iff c \sim a \wedge c \sim b \iff a \sim b,$$

using symmetry and transitivity for the last equivalence. So $\Psi(\Phi(\sim)) = \sim$.

The two maps are mutually inverse, hence bijective. $\square$

3 An Example to Hold Onto

Example 1 (Congruence mod n)**.** Fix $n \in \mathbb{Z}_{>0}$. Define $a \equiv b \pmod{n}$ iff $n \mid (a - b)$. This is an equivalence relation on $\mathbb{Z}$, and Theorem **??** shows the residue classes $[0], [1], \ldots, [n-1]$ form a partition of $\mathbb{Z}$ into exactly n blocks.

Example 2 (Divisibility)**.** On $\mathbb{N}_{>0}$, define $a \leq_{\mid} b$ iff $a \mid b$. This is a partial order: reflexive ($a \mid a$), antisymmetric ($a \mid b \wedge b \mid a \Rightarrow a = b$ for positive integers), and transitive. It is *not* total: $2 \nmid 3$ and $3 \nmid 2$.